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sh mpls int
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R2(config)#
*Sep  3 05:08:24.367: %LDP-5-NBRCHG: LDP Neighbor 3.3.3.3:0 (1) is UP
R2(config)#end
R2#sh mpls
*Sep  3 05:08:54.951: %SYS-5-CONFIG_I: Configured from console by consoleint
Interface          IP          Tunnel    BGP  Static Operational
GigabitEthernet0/0  Yes (ldp)   No        No   No        Yes
GigabitEthernet0/1  Yes (ldp)   No        No   No        Yes
R2#
```

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show mpls ldp neighbor
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```
R2#show mpls ldp neighbor
  Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 2.2.2.2:0
    TCP connection: 3.3.3.3.56461 - 2.2.2.2.646
    State: Oper; Msgs sent/rcvd: 11/11; Downstream
    Up time: 00:03:05
    LDP discovery sources:
      GigabitEthernet0/1, Src IP addr: 10.0.23.2
    Addresses bound to peer LDP Ident:
      10.0.23.2      10.0.34.1      3.3.3.3
R2#
```

```
show mpls ldp bindings
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R2#show mpls ldp bindings
  lib entry: 2.2.2.2/32, rev 2
    local binding:  label: imp-null
    remote binding: lsr: 3.3.3.3:0, label: 16
  lib entry: 3.3.3.3/32, rev 4
    local binding:  label: 16
    remote binding: lsr: 3.3.3.3:0, label: imp-null
  lib entry: 10.0.12.0/30, rev 6
    local binding:  label: imp-null
    remote binding: lsr: 3.3.3.3:0, label: 17
  lib entry: 10.0.23.0/30, rev 8
    local binding:  label: imp-null
    remote binding: lsr: 3.3.3.3:0, label: imp-null
  lib entry: 10.0.34.0/30, rev 10
    local binding:  label: 17
    remote binding: lsr: 3.3.3.3:0, label: imp-null
R2#
```

```
show mpls forwarding-table
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```

R2#show mpls forwarding-table
Local      Outgoing  Prefix          Bytes Label  Outgoing  Next Hop
Label      Label      or Tunnel Id    Switched     interface
16         Pop Label   3.3.3.3/32      0           Gi0/1      10.0.23.2
17         Pop Label   10.0.34.0/30    0           Gi0/1      10.0.23.2
R2#

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On R1,

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sh ip bgp
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R1#sh ip bgp
BGP table version is 3, local router ID is 172.16.1.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*>  172.16.1.0/24    0.0.0.0             0         32768 i
*>  172.16.4.0/24    10.0.12.2           0         65000 65020 i
R1#

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ping 172.16.4.1 (from R1 to R4)

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R1#ping 172.16.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.4.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/3/3 ms
R1#

```